

ESAT Biology

Formula + Things to Remember

A compact, exam-facing sheet for formulae, ratios, definitions, process order, graph facts and high-yield biological recall.

ESAT Biology style: fast multiple choice rewards precise definitions, process order, rate/graph reasoning and avoiding common misconception traps.

1. Cells, movement and enzymes

ESAT SCOPE MARKERS

- core** Only content from Biology B1-B11 is included.
- useful skill** Formula/rate/graph shortcuts are included where they help ESAT multiple choice.
- trap** This is not a full textbook summary; it is a recall and calculation sheet.

EUKARYOTIC CELL COMPONENTS

- Cell membrane: partially permeable boundary; controls exchange.
- Cytoplasm: site of many chemical reactions.
- Nucleus: contains DNA/chromosomes; controls cell activities.
- Mitochondria: aerobic respiration and ATP release.
- Plant wall: cellulose support; chloroplast: photosynthesis; vacuole: cell sap/turgor.

PROKARYOTIC CELL COMPONENTS

- Cell membrane, cytoplasm and cell wall.
- Chromosomal DNA is free in cytoplasm; no true nucleus.
- Plasmids are small extra loops of DNA.
- trap** Prokaryotic DNA is not inside a nucleus.

ORGANISATION HIERARCHY

cells → tissues → organs → organ systems → organism
Tissue = group of similar cells with a shared function. Organ = different tissues working together.

DIFFUSION

net movement: high concentration → low concentration
Passive; no ATP directly required.
rate ∝ surface area × concentration difference / diffusion distance
shortcut Large SA, thin barrier, steep gradient, high temperature = faster.

OSMOSIS

Net movement of water across a partially permeable membrane from higher water potential to lower water potential.
trap Phrase using water potential, not just “water moves to more solute”.

ACTIVE TRANSPORT

Movement against a concentration gradient using ATP and carrier proteins.
Examples: mineral ions into root hair cells; glucose uptake in gut/kidney.
must Anything needing active transport depends on respiration for ATP.

SURFACE AREA TO VOLUME

SA:V = surface area / volume
Small cells/organisms exchange efficiently; larger organisms need exchange surfaces and transport systems.
trap If length scale doubles: SA ×4, volume ×8, SA:V halves.

ENZYME BASICS

Enzymes are proteins and biological catalysts. Active site is specific to substrate.
 $E + S \rightleftharpoons ES \text{ complex} \rightarrow E + P$
They lower activation energy and are not used up.

ENZYME RATE GRAPHS

- Temperature: rises to optimum, then sharp fall as enzyme denatures.
- pH: optimum pH; extremes change active site shape.
- Substrate concentration: rises then plateaus when active sites are saturated.
- Low temperature slows collisions but does not denature.

DIGESTIVE ENZYMES

Enzyme	Substrate	Product
Amylase	starch	sugars
Protease	proteins	amino acids
Lipase	lipids	fatty acids + glycerol

2. Cell division, inheritance, DNA and gene technologies

MITOSIS

Produces two genetically identical diploid daughter cells for growth, repair, cell replacement and asexual reproduction.
DNA replicates → chromosomes separate → two identical cells

trap Cancer = mutations causing uncontrolled cell division.

MEIOSIS

Produces four genetically different haploid gametes after two divisions.
Halves chromosome number; variation arises from chromosome mixing and fertilisation.

trap Meiosis is for gametes, not body-cell growth.

ASEXUAL VS SEXUAL REPRODUCTION

Type	Parents	Genetic outcome
Asexual	one	clones, low variation
Sexual	two/gametes	variation from allele mixing

SEX DETERMINATION

Female: XX. Male: XY.
Egg always carries X; sperm carries X or Y.
XX = female, XY = male; expected ratio about 1:1.

INHERITANCE VOCABULARY

- Gene: DNA section controlling a characteristic/polypeptide.
- Allele: version of a gene.
- Genotype: alleles present; phenotype: expressed characteristic.
- Dominant: expressed with one copy; recessive: needs two copies.
- Homozygous: same alleles; heterozygous: different alleles.
- Autosome: non-sex chromosome.

MONOHYBRID RATIOS

$Aa \times Aa \rightarrow 1 AA : 2 Aa : 1 aa$
If A dominant: phenotype ratio 3 dominant : 1 recessive.
Test cross: unknown dominant phenotype $\times$ homozygous recessive.

PROBABILITY IN GENETICS

Independent events: multiply probabilities.
Alternative routes: add probabilities.

trap Previous child does not change probability for the next child.

DNA ESSENTIALS

- Genome = full genetic material of an organism.
- DNA is a double helix made of nucleotides.
- Nucleotide = sugar + phosphate + base.
- Complementary base pairs: A-T and C-G.
- Triplets of bases code for amino acids.

FROM GENE TO PROTEIN

gene base sequence → amino-acid sequence → polypeptide → protein
A mutation changes nucleotide sequence; effect may be none, small or major.

GENETIC ENGINEERING SEQUENCE

isolate useful gene → cut DNA/vector → insert gene → ligase joins → organism expresses gene
Restriction enzymes cut DNA; ligase joins DNA fragments.
Examples include medical protein production and crop traits.

STEM CELLS

Type	Potential
Totipotent	can form all cell types + extra-embryonic tissue
Pluripotent	can form many body cell types
Multipotent	limited range, e.g. adult stem cells

SELECTIVE BREEDING VS NATURAL SELECTION

Selective breeding: humans choose parents with desired traits.
Natural selection: environment creates selection pressure; advantageous alleles become more common.

trap Selection acts on phenotype; allele frequencies change over generations.

3. Animal physiology and control systems

RESPIRATION EQUATIONS

aerobic: glucose + oxygen → carbon dioxide + water + ATP
 anaerobic in animals: glucose → lactic acid + less ATP
 Extra oxygen after exercise helps break down lactic acid and restore normal stores.

GAS EXCHANGE SURFACES

- Large surface area.
- Short diffusion distance: thin walls.
- Steep concentration gradient maintained by ventilation/blood flow.
- Moist surface for gases to dissolve.

RESPIRATORY SYSTEM REMINDERS

Ventilation moves air in/out of lungs; gas exchange happens at alveoli.
 O₂ diffuses alveolus → blood; CO₂ diffuses blood → alveolus.
trap Direction depends on concentration/partial-pressure gradient.

CIRCULATORY FORMULAS

cardiac output = heart rate × stroke volume
 Arteries: high pressure, thick elastic walls. Veins: valves, low pressure. Capillaries: one-cell-thick exchange surfaces.

BLOOD COMPONENTS

- Red blood cells: haemoglobin, no nucleus, biconcave, carry oxygen.
- Plasma: transports CO₂, urea, hormones, heat and soluble products.
- Platelets: clotting.
- White blood cells: immune defence.

NERVOUS PATHWAY

receptor → sensory neurone → CNS → motor neurone → effector
 Synapses use neurotransmitter across a gap; one-way transmission.
 Reflex arcs are rapid, automatic and protective.

NEGATIVE FEEDBACK

stimulus → receptor → coordination centre → effector → response
 Response reverses deviation from set point.
shortcut If level is too high, correction lowers it; if too low, correction raises it.

BLOOD GLUCOSE CONTROL

High glucose: pancreas releases insulin → cells take up glucose; liver stores glucose as glycogen.
 Low glucose: pancreas releases glucagon → liver converts glycogen to glucose.
trap Insulin lowers; glucagon raises.

DIABETES

Type 1: pancreas produces little/no insulin; treated with insulin and blood-glucose management.
 Type 2: cells become less responsive to insulin; associated with lifestyle/genetic risk factors.

THERMOREGULATION

Too hot	Too cold
vasodilation	vasoconstriction
sweating	shivering
hairs lie flat	hairs raised

KIDNEY AND WATER CONTROL

Nephrons filter blood, then selectively reabsorb useful substances.
 ADH increases water reabsorption in kidneys; urine becomes more concentrated.
trap More ADH means less water lost in urine.

MENSTRUAL CYCLE HORMONES

- FSH: stimulates follicle/egg maturation and oestrogen release.
- Oestrogen: rebuilds uterus lining; inhibits FSH; stimulates LH surge.
- LH: triggers ovulation.
- Progesterone: maintains uterus lining; inhibits FSH/LH.

4. Disease, defence, ecosystems and cycles

PATHOGENS

Type	Key idea
bacteria	living cells; may release toxins
viruses	replicate inside host cells
protists	single-celled eukaryotes; some parasitic
fungi	can cause plant/animal disease

DEFENCES AND IMMUNITY

- Skin/clotting, mucus/cilia and stomach acid are first-line barriers.
- Phagocytes engulf pathogens.
- Lymphocytes produce specific antibodies.
- Memory cells make secondary response faster and stronger.

ANTIBIOTICS AND VACCINES

Antibiotics kill bacteria, not viruses.
Vaccines introduce antigen safely to produce memory cells.
Antibiotic resistance evolves by natural selection after variation/mutation.

MEDICINE/VACCINE DEVELOPMENT

Preclinical testing checks cells/tissues/animals before humans.
Clinical trials check safety, dose and effectiveness.
Placebos, blind/double-blind trials and large samples reduce bias.

NONCOMMUNICABLE DISEASE

Risk factors may be lifestyle, genetic or environmental.
CVD management: lifestyle change, statins, anticoagulants, antihypertensives, stents or bypass surgery.
Cancer involves uncontrolled cell division; benign vs malignant tumours.

ECOLOGY VOCABULARY

- Individual: one organism.
- Population: one species in an area.
- Community: all populations in an area.
- Ecosystem: community plus abiotic environment.
- Abiotic = non-living factor; biotic = living factor.

POPULATION INTERACTIONS

- Competition: organisms require same limited resource.
- Predation: predator benefits, prey harmed.
- Mutualism: both species benefit.
- Parasitism: parasite benefits, host harmed.

PRODUCERS AND BIOMASS

Photosynthetic producers transfer light energy into chemical energy in biomass.
Biomass decreases up food chains due to respiration, heat loss, movement, waste, uneaten parts and egestion.
$$\text{efficiency} = \frac{\text{useful energy transferred}}{\text{total input}} \times 100\%$$

CARBON CYCLE QUICK

CO₂ removed by photosynthesis
CO₂ released by respiration, combustion and decomposition
trap Energy flows one way; matter cycles.

WATER CYCLE QUICK

evaporation/transpiration → condensation → precipitation → runoff/infiltration
Plants contribute by transpiration.

SAMPLING

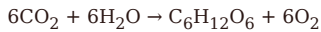
population estimate = mean per quadrat × total area / quadrats
Use random quadrats to reduce bias; use belt transects across gradients.

HUMAN ECOSYSTEM IMPACTS

- Fish farming can affect waste, disease, parasites and local food webs.
- Acid rain damages organisms and habitats.
- Eutrophication: excess nutrients → algal bloom → light blocked → decay uses oxygen → aquatic organisms die.

5. Plant physiology and final data shortcuts

PHOTOSYNTHESIS EQUATION



Requires light energy and chlorophyll; it is endothermic.
Glucose can be used for respiration, starch, cellulose, amino acids and lipids.

LIMITING FACTORS

- Light intensity: more photons until another factor limits.
- CO₂ concentration: more substrate until saturation.
- Temperature: enzyme-controlled reactions rise then fall if enzymes denature.
- Chlorophyll amount affects light absorption.

LIGHT INTENSITY SHORTCUT

$$\text{light intensity} \propto 1 / \text{distance}^2$$

Double distance → one-quarter intensity. Triple distance → one-ninth.

XYLEM

Dead, hollow, lignified tubes carry water and mineral ions from roots to leaves.

Movement is driven by transpiration pull and cohesion between water molecules.

PHLOEM

Living tissue transports dissolved sugars and amino acids by translocation from source to sink.

Requires ATP for loading/unloading.

trap Phloem can move up or down; xylem mainly upward.

ROOT HAIR CELLS

Large surface area for absorption.

Mineral ions often enter by active transport; water follows by osmosis.

Many mitochondria supply ATP for active uptake.

STOMATA AND GUARD CELLS

Stomata allow CO₂ entry but water vapour loss.

Guard cells turgid → pore opens; flaccid → closes.

Light usually opens stomata for photosynthesis.

TRANSPIRATION

$$\text{rate} = \text{volume of water lost} / \text{time}$$

Rate increases with light, temperature and air movement; decreases with high humidity.

trap High humidity lowers water-potential gradient, so transpiration slows.

RATE AND PERCENTAGE

$$\text{rate} = \text{change} / \text{time}$$

$$\% \text{ change} = (\text{new} - \text{old}) / \text{old} \times 100$$

Use original value in denominator.

GRAPH INSTINCTS

- Gradient = rate.
- Plateau = limiting factor, substrate used up, maximum reached or equilibrium.
- Independent variable on x-axis; dependent on y-axis.
- Correlation does not prove causation.
- Anomaly = inconsistent result; repeat or exclude with reason.

MOST COMMON BIOLOGY TRAPS

- Mitosis makes identical diploid cells; meiosis makes varied haploid gametes.
- Antibiotics do not kill viruses.
- Active transport needs ATP; diffusion/osmosis do not directly need ATP.
- Water potential wording matters in osmosis.
- Insulin lowers blood glucose; glucagon raises it.
- Energy flows; carbon/water cycle.